

Introduction

The Online Voting System is a secure, web-based application designed to facilitate electronic voting with robust user authentication via facial recognition. Built using Python and leveraging a suite of modern libraries, the system ensures a user-friendly interface, secure data management, and reliable vote tracking. This document provides an in-depth technical breakdown of the project's components, based on the provided Python code, covering the technologies, implementation details, and key features. It includes additional technical insights into the codebase, database operations, facial recognition algorithms, and deployment considerations, aiming to provide a comprehensive understanding for developers, system architects, and stakeholders.

1. Frontend

Overview

The frontend is the user interface, enabling voters to register, log in, cast votes, view results, and manage profiles. It is built using **Streamlit**, a Python-based framework that abstracts traditional web development (HTML, CSS, JavaScript) into Python, allowing rapid development of interactive applications.

Implementation Details

- **Streamlit Framework:**
 - Streamlit version 1.x (assumed based on syntax) is used to create a single-page application with a reactive UI.
 - Key components include:
 - `st.text_input` and `st.selectbox` for form inputs (e.g., username, candidate selection).
 - `st.form` for grouped input submission, reducing unnecessary reruns (Streamlit's default behavior is to rerun the script on every interaction).
 - `st.button` and `st.checkbox` for user actions (e.g., vote submission, vote confirmation).
 - `st.tabs` to organize the interface into sections: **Home**, **Vote**, **Results**, **Profile**, and **Admin** (admin-only).
 - `st.expander` for collapsible candidate descriptions, optimizing screen space.

- `st.image` for real-time webcam frame display during facial recognition.
 - The `st.session_state` object manages state persistence across reruns, storing:
 - `user_id` : Integer or None, representing the logged-in user's ID.
 - `has_voted` : Boolean, tracking voting status.
 - `username` : String, storing the logged-in user's username.
 - `is_admin` : Boolean, indicating admin privileges.
 - `page` : String, controlling navigation (e.g., "welcome", "register", "dashboard").
 - `register_data` and `login_data` : Dictionaries for multi-step form data.
 - The `st.set_page_config(page_title="Online Voting System", page_icon=" ", layout="wide")` sets a wide layout and custom branding.
- **Custom Styling:**
 - CSS is injected via `st.markdown` with `unsafe_allow_html=True`, bypassing Streamlit's default styling.
 - The dark theme uses a gradient background(`linear-gradient(135deg, #1a1a1a, #2d2d2d)`), white text(`color: #ffffff`), and a primary color(`#00aaff`) for buttons and accents.
 - Buttons feature:
 - Hover effects(`transition: all 0.3s ease; background-color: #0088cc`).
 - Scaling animations(`transform: scale(1.05)`).
 - Rounded corners(`border-radius: 8px`).
 - Form sections use `box-shadow: 0 4px 10px rgba(0,0,0,0.5)` for depth and `padding: 20px` for spacing.
 - The `.webcam-container` and `.countdown` classes style the facial recognition UI, centering the webcam feed and timer.
 - **Dynamic Features:**
 - **Webcam Integration:** OpenCV(`cv2.VideoCapture(0)`) captures webcam frames, resized to 640x480 pixels for performance. Frames are converted from BGR to RGB (`cv2.cvtColor(frame, cv2.COLOR_BGR2RGB)`) for Streamlit's `st.image`.
 - **Real-Time Feedback:** The `st.empty` placeholder updates the UI with countdown timers (`<div class="countdown">`) and warnings(`st.warning` , `st.error`) during face capture.
 - **Data Visualization:** The `get_results` function uses `pandas.DataFrame` to structure vote data, visualized with `st.bar_chart` for vote counts and `st.metric` for vote percentages.
 - **Responsive Design:** The `st.columns` layout organizes content (e.g., username display and logout button), with a 0.8:0.2 ratio for alignment.

Technical Insights

- **Performance:** Streamlit's reactive model reruns the script on every interaction, which can be mitigated by using `st.form` to batch inputs and `@st.cache_data` for expensive operations (e.g., `get_candidates`).
- **Challenges:** The `unsafe_allow_html=True` approach for CSS injection is brittle, as Streamlit may deprecate or restrict it in future versions. A component-based styling solution (e.g., Streamlit's `components.v1.html`) could be more robust.
- **Scalability:** Streamlit is single-threaded by default, limiting concurrent users. For production, deploying with `streamlit run --server.enableCORS false` or a multi-threaded server (e.g., Gunicorn) is necessary.

Significance

- Streamlit's Python-centric approach reduces frontend complexity, enabling developers to focus on functionality.
 - Custom styling builds trust through a professional UI, critical for a voting system.
 - Dynamic features like webcam integration enhance user engagement, particularly for biometric authentication.
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2. Backend

Overview

The backend manages data storage, user authentication, vote processing, and administrative tasks using **SQLite** and Python functions, ensuring secure and efficient operations.

Implementation Details

- **SQLite Database:**
 - SQLite, a C-based, serverless database, stores data in `election.db`.
 - The `init_db` function creates three tables:
 - **users:**

```
CREATE TABLE IF NOT EXISTS users (  
    id INTEGER PRIMARY KEY AUTOINCREMENT,  
    username TEXT UNIQUE,  
    password TEXT,  
    full_name TEXT,
```

```

        gender TEXT,
        has_voted BOOLEAN,
        face_encoding_path TEXT
    )

```

■ **candidates:**

```

CREATE TABLE IF NOT EXISTS candidates (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    name TEXT UNIQUE,
    description TEXT,
    votes INTEGER
)

```

■ **votes:**

```

CREATE TABLE IF NOT EXISTS votes (
    id INTEGER PRIMARY KEY AUTOINCREMENT,
    user_id INTEGER,
    candidate_id INTEGER,
    timestamp TEXT,
    FOREIGN KEY(user_id) REFERENCES users(id),
    FOREIGN KEY(candidate_id) REFERENCES candidates(id)
)

```

- Foreign keys in `votes` ensure referential integrity, preventing invalid user or candidate IDs.
- Database operations use `sqlite3.connect` with a cursor(`conn.cursor()`) for queries, committing changes with `conn.commit()` and closing connections with `conn.close()`.

• **File Storage:**

- **Face Encodings:** The `face_encodings` directory stores 128-dimensional face encodings as `.npy` files(e.g., `<username>.npy`) using `np.save`. Each encoding is a NumPy array of floats, typically ~1KB in size.
- **Database Backups:** The `reset_votes` function uses `shutil.copy` to create timestamped backups(e.g., `election_backup_20250421_143022.db`) before resetting votes.

• **Business Logic:**

- **User Management:**
 - `register_user`: Validates inputs, hashes passwords, generates face encodings, and inserts user data into `users`.
 - `login_user`: Verifies credentials and face encodings, returning user ID and voting status.

- `update_user_profile` : Updates `username` , `full_name` , and `gender` with validation for unique usernames.
- **Voting Logic:**
 - `cast_vote` : Checks `has_voted` , updates `users.has_voted` , increments `candidates.votes` , and logs the vote in `votes` with `datetime.now().isoformat()` .
 - Uses transactions (`conn.commit()`) to ensure atomicity.
- **Admin Functions:**
 - `add_candidate` : Inserts a candidate with a unique name and optional description, handling `sqlite3.IntegrityError` for duplicates.
 - `reset_votes` : Resets `candidates.votes` , `users.has_voted` , and deletes `votes` entries after backing up the database.
- **Data Retrieval:**
 - `get_candidates` : Fetches all candidates with `@st.cache_data` for performance.
 - `get_results` : Aggregates vote counts and calculates percentages using list comprehension.

Technical Insights

- **Database Performance:** SQLite uses a single file, limiting concurrency. The `WAL` (Write-Ahead Logging) mode could be enabled (`PRAGMA journal_mode=WAL`) for better performance under load.
- **Error Handling:** Try-except blocks catch `sqlite3.IntegrityError` for unique constraint violations and generic exceptions for file operations, ensuring robustness.
- **File Storage:** The `.npy` format is efficient for NumPy arrays but lacks encryption. Storing encodings in a secure database column (e.g., BLOB) with AES encryption could enhance security.
- **Atomicity:** Database transactions are atomic, but long-running operations (e.g., `reset_votes`) could benefit from explicit transaction management (`BEGIN TRANSACTION`).

Significance

- SQLite's lightweight nature suits small-scale deployments, requiring no server setup.
- Modular functions enhance maintainability, allowing easy extension (e.g., adding audit logs).
- File-based storage ensures data persistence, critical for vote integrity and recovery.

3. Programming Language

Overview

The application is written entirely in **Python 3.x**, leveraging its readability, extensive libraries, and cross-platform compatibility.

Implementation Details

- Python handles both frontend (via Streamlit) and backend logic, using:
 - **Standard Libraries:** `sqlite3`, `hashlib`, `re`, `datetime`, `os`, `time`, `logging`.
 - **Third-Party Libraries:** `streamlit`, `face_recognition`, `opencv-python`, `numpy`, `pandas`, `shutil`.
- The codebase uses functional programming with modular functions (e.g., `hash_password`, `get_face_encoding`) and minimal global state, managed via `st.session_state`.
- Error handling employs try-except blocks, logging exceptions with `logging.error` for debugging.
- The `if __name__ == "__main__": main()` structure ensures the application runs only when executed directly.

Technical Insights

- **Version Compatibility:** The code assumes Python 3.7+ (due to `face_recognition` and Streamlit requirements). Explicit version checks (`sys.version_info`) could ensure compatibility.
- **Performance:** Python's interpreted nature is sufficient for this application but may bottleneck under high load. Cython or NumPy optimizations could speed up face recognition.
- **Logging:** The `logging.basicConfig(level=logging.DEBUG)` configuration logs to stdout. For production, a file-based logger (`FileHandler`) with rotation (`RotatingFileHandler`) would be more robust.

Significance

- Python's ecosystem enables complex features (e.g., facial recognition, web UI) in a single language, reducing development complexity.
 - Its cross-platform nature supports deployment on diverse systems.
 - The modular design facilitates maintenance and future enhancements.
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4. Security

Overview

Security is paramount, with features like password hashing, input validation, facial recognition, and admin access controls to protect user data and vote integrity.

Implementation Details

- **Password Hashing:**

- The `hash_password` function uses SHA-256 (`hashlib.sha256`), producing a 64-character hexadecimal digest.
- Example: `hashlib.sha256("password".encode()).hexdigest()` yields a secure hash.
- Passwords are stored in the `users.password` column, never in plaintext.

- **Input Validation:**

- `is_valid_username` uses `regex( re.match )`:
 - Alphanumeric: `^[a-zA-Z0-9]{3,20}$`
 - Email: `^[w\.-]+@[w\.-]+\.\w+$`
 - Mobile: `^\+?\d{10,15}$`
- `is_valid_password` enforces a minimum length of 6 characters.
- Validation prevents SQL injection and invalid data.

- **Facial Recognition:**

- The `face_recognition` library, built on `dlib`, uses a pre-trained deep learning model (HOG-based face detector + CNN for encodings).
- The `capture_face` function:
 - Captures frames at 640x480 resolution.
 - Uses `face_recognition.face_locations(model="small")` for faster detection.
 - Generates a 128-dimensional encoding with `face_recognition.face_encodings`.
 - Saves encodings as `.npy` files.
- The `verify_face` function:
 - Compares live encodings against stored ones using `face_recognition.compare_faces` with a tolerance of 0.4 (Euclidean distance threshold).
 - Runs for up to 30 seconds, with a 0.1-second sleep (`time.sleep(0.1)`) to reduce CPU usage.
 - Handles edge cases (multiple faces, no faces) with warnings.

- **Admin Authentication:**

- The `is_admin` function checks for `username == "admin" and password == "admin123"`.
- Admins access the **Admin** tab for candidate management and vote resets.
- **Data Protection:**
 - Face encodings are stored in `face_encodings`, with file-level access control assumed.
 - The UI claims encryption, but no explicit encryption (e.g., AES) is implemented in the code.
- **Vote Integrity:**
 - The `cast_vote` function enforces one vote per user via `users.has_voted`.
 - The `is_voting_open` function checks `datetime.now() <= datetime(2025, 12, 31, 23, 59, 59)`.

Technical Insights

- **SHA-256 Limitations:** SHA-256 is secure but lacks salting, making it vulnerable to rainbow table attacks. Using `bcrypt` or `argon2` with salts would be more secure.
- **Facial Recognition:**
 - The “small” HOG model is CPU-efficient but less accurate than CNN-based models. Switching to `model="cnn"` with GPU support could improve accuracy at the cost of performance.
 - The 0.4 tolerance is empirical; tuning it (e.g., 0.3 for stricter matching) requires testing across diverse conditions.
- **SQL Injection:** Parameterized queries (`c.execute("SELECT ... WHERE username = ?", (username,))`) prevent injection, but input sanitization could be extended to other fields.
- **Admin Security:** Hardcoded credentials are a critical flaw. Storing admin credentials in a hashed `admins` table or using OAuth would mitigate risks.

Significance

- Password hashing and validation provide a baseline for secure authentication.
- Facial recognition adds a biometric layer, reducing impersonation risks.
- Vote integrity ensures a fair election process, though admin security needs improvement.

5. Libraries and Dependencies

Overview

The project uses a mix of Python standard and third-party libraries to implement its features.

Implementation Details

- **Streamlit**: Version 1.x, for the web UI, leveraging `st.*` APIs for components and state management.
- **sqlite3**: Built-in, for database operations with `sqlite3.connect` and cursor-based queries.
- **hashlib**: Provides SHA-256 hashing via `hashlib.sha256`.
- **re**: Supports regex validation with `re.match`.
- **datetime**: Manages timestamps (`datetime.now().isoformat()`) and deadlines.
- **pandas**: Version 1.x, for DataFrame-based vote aggregation and visualization (`st.bar_chart`).
- **shutil**: Handles file copying for database backups (`shutil.copy`).
- **OpenCV (cv2)**: Version 4.x, for webcam access (`cv2.VideoCapture`) and image processing (`cv2.resize`, `cv2.cvtColor`).
- **face_recognition**: Version 1.x, built on `dlib`, for face detection and encoding.
- **NumPy**: Version 1.x, for array operations (`np.save`, `np.load`) and face encodings.
- **os**: Manages directories (`os.makedirs`, `os.path.exists`) and file paths.
- **time**: Implements timers (`time.time`, `time.sleep`) for face capture and verification.
- **logging**: Configured with `logging.basicConfig(level=logging.DEBUG)` for detailed logs.

Technical Insights

- **Dependency Management**: A `requirements.txt` file (e.g., `streamlit==1.28.0`, `face_recognition==1.3.0`) ensures reproducible environments. `pipdeptree` can audit dependencies for conflicts.
- **face_recognition**: Requires `dlib`, which needs CMake and a C++ compiler, complicating installation on some systems. Pre-built wheels (`pip install dlib --verbose`) can simplify setup.
- **OpenCV**: The `cv2.VideoCapture(0)` assumes the default webcam. Supporting multiple cameras (`cv2.VideoCapture(1)`) or fallbacks requires additional logic.

Significance

- The library ecosystem enables complex functionality with minimal custom code.
- `logging` aids debugging, critical for facial recognition and database issues.
- Dependency complexity (e.g., `dlib`) requires clear installation instructions.

6. Key Features

Overview

The system offers user registration, login with facial recognition, voting, result visualization, profile management, and admin controls.

Implementation Details

- **User Registration:**

- The `register_form` collects `username`, `password`, `full_name`, and `gender` (`st.selectbox` with `["Male", "Female", "Other", "Prefer not to say"]`).
- `capture_face` uses `cv2.VideoCapture` to capture a frame, resized to 640x480, and processed with `face_recognition.face_locations`.
- `register_user` validates inputs, generates a face encoding, saves it as `<username>.npy`, and inserts data into `users`.

- **User Login:**

- The `login_form` collects `username` and `password`.
- `login_user` queries `users` for credentials, loads the stored encoding (`np.load`), and calls `verify_face`.
- `verify_face` captures live frames, generates encodings, and compares them with a 0.4 tolerance.

- **Voting:**

- The **Vote** tab uses `st.expander` for candidate details and `st.selectbox` for selection.
- `cast_vote` executes:

```
UPDATE users SET has_voted = ? WHERE id = ?
UPDATE candidates SET votes = votes + 1 WHERE id = ?
INSERT INTO votes (user_id, candidate_id, timestamp) VALUES (?, ,
```

- Voting is disabled if `has_voted` is True or post-deadline.

- **Results Visualization:**

- `get_results` computes vote percentages: `[(name, votes, votes / total_votes * 100) for name, votes in candidates]`.
- Displays metrics (`st.metric`) and a bar chart (`st.bar_chart(df.set_index("Candidate")["Votes"])`).

- **Profile Management:**

- `get_user_profile` fetches `username`, `full_name`, `gender`, `has_voted` from `users`.
- `update_user_profile` updates `users` with:

```
UPDATE users SET username = ?, full_name = ?, gender = ? WHERE id = ?
```

- **Admin Panel:**

- `add_candidate` inserts into `candidates` with `votes = 0`.
- `reset_votes` executes:

```
UPDATE candidates SET votes = 0
UPDATE users SET has_voted = 0
DELETE FROM votes
```

- Requires "CONFIRM" input for safety.

- **Facial Recognition:**

- Uses `face_recognition.face_locations(model="small")` for detection and `face_recognition.face_encodings` for 128-dimensional encodings.
- `capture_face` auto-captures within 5 seconds; `verify_face` runs for 30 seconds with real-time feedback.

Technical Insights

- **Facial Recognition Performance:** The HOG model processes ~10 FPS on a mid-range CPU. GPU acceleration (via `dlib`'s CUDA support) could improve speed.
- **Database Queries:** `get_candidates` uses `SELECT *`, which could be optimized to `SELECT id, name, description, votes` for clarity.
- **UI Reruns:** Streamlit's reruns on input changes can cause flickering during face capture. Using `st.form` for face verification could stabilize the UI.

Significance

- Facial recognition enhances security, making the system modern and trustworthy.
 - Voting and result features ensure transparency and fairness.
 - The admin panel provides efficient election management.
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7. Deployment Considerations

Overview

Deploying the system requires addressing environment setup, dependencies, and hardware constraints, particularly for webcam-based facial recognition.

Implementation Details

- **Environment:**

- Runs with `streamlit run <script_name>.py`, defaulting to `http://localhost:8501`.
- Supports Linux, macOS, Windows with Python 3.7+.

- **Dependencies Installation:**

- Requires: `streamlit`, `face_recognition`, `opencv-python`, `numpy`, `pandas`, `shutil`.
- `face_recognition` installation:

```
pip install dlib --verbose
pip install face_recognition
```

- A `requirements.txt` is recommended:

```
streamlit==1.28.0
face_recognition==1.3.0
opencv-python==4.8.0
numpy==1.24.0
pandas==2.0.0
```

- **Webcam Dependency:**

- `cv2.VideoCapture(0)` assumes a default webcam. Multi-camera support requires dynamic device selection.
- Cloud platforms (e.g., Heroku) lack webcam access, necessitating local or hardware-equipped servers (e.g., Raspberry Pi with a camera).

- **Caching:**

- `@st.cache_data` on `get_candidates` hashes query results, reducing database load.
- Cache invalidation occurs on database changes, requiring careful management.

- **Logging:**

- Logs are output to stdout. For production, configure:

```
logging.basicConfig(filename="app.log", level=logging.INFO, form
```

Technical Insights

- **Scaling:** Streamlit's single-threaded server limits concurrent users. Use `gunicorn` with multiple workers (`gunicorn -w 4 streamlit_app:app`) for scalability.

- **Webcam Alternatives:** Client-side webcam access via JavaScript (e.g., `navigator.mediaDevices.getUserMedia`) could enable cloud deployment but requires rewriting the face capture logic.
- **Monitoring:** Integrate Prometheus or ELK for log aggregation and performance monitoring in production.

Significance

- Local deployment suits small-scale elections (e.g., schools, organizations).
 - Dependency management ensures reproducibility, though `dlib` setup needs documentation.
 - Webcam constraints highlight the need for strategic deployment planning.
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8. Project Structure

Overview

The project is a single Python script, combining frontend and backend logic for simplicity.

Implementation Details

- **Single File:**
 - The script (e.g., `main.py`) contains ~600 lines, including UI, logic, and utilities.
 - The `main` function initializes the database (`init_db`), sets up `st.session_state`, and renders the UI based on `page`.
- **Session State:**
 - `st.session_state` stores:

```
st.session_state.user_id = None
st.session_state.has_voted = False
st.session_state.username = None
st.session_state.is_admin = False
st.session_state.page = "welcome"
st.session_state.register_data = None
st.session_state.login_data = None
```

- State persists across reruns, enabling multi-step workflows (e.g., registration → face capture).
- **Modular Functions:**
 - Organized into categories:

- Database: `init_db`, `register_user`, `login_user`, `cast_vote`, `add_candidate`, `reset_votes`, `update_user_profile`.
- Face Recognition: `capture_face`, `verify_face`, `get_face_encoding`.
- Validation: `is_valid_username`, `is_valid_password`, `hash_password`.
- Utilities: `get_candidates`, `get_results`, `get_user_profile`, `is_admin`, `is_voting_open`.
- Functions are stateless, relying on parameters and database state.
- **Directory Structure:**
 - `face_encodings/` : Stores `.npy` files.
 - `election.db` : SQLite database.
 - Backups: `election_backup_<timestamp>.db`.

Technical Insights

- **Refactoring Potential:** Splitting into modules (e.g., `database.py`, `face_recognition.py`, `ui.py`) would improve maintainability for larger teams.
- **State Management:** `st.session_state` is effective but lacks type safety. A typed dictionary (`typing.TypedDict`) could enforce structure.
- **File Growth:** The single-file approach may become unwieldy; a Flask or FastAPI backend with Streamlit as a frontend could modularize the architecture.

Significance

- The single-file structure simplifies development and deployment for small projects.
- Modular functions support extensibility, e.g., adding two-factor authentication.
- Session state ensures a seamless user experience.

9. Limitations

Overview

The system has limitations that impact security, scalability, and usability, requiring attention for broader adoption.

Implementation Details

- **Admin Security:**

- Hardcoded credentials(`is_admin: username == "admin" and password == "admin123"`) are vulnerable to brute-force attacks.
- Mitigation: Use a `admins` table with hashed passwords or OAuth integration.
- **Facial Recognition:**
 - The HOG model(`model="small"`) is less accurate under poor lighting or with occlusions (e.g., masks).
 - The 0.4 tolerance may cause false negatives; dynamic tuning based on user feedback could help.
 - The 5-second capture and 30-second verification timeouts are fixed, potentially frustrating users.
- **Scalability:**
 - SQLite's file-based locking limits concurrency. For >100 users, a client-server database (e.g., PostgreSQL) is needed.
 - Streamlit's single-threaded model requires a multi-worker setup for high traffic.
- **Hardcoded Deadline:**
 - The deadline(`datetime(2025, 12, 31, 23, 59, 59)`) is hardcoded, requiring code changes.
 - A database-stored deadline or environment variable (`os.getenv("VOTING_DEADLINE")`) would be flexible.
- **Error Handling:**
 - Webcam failures(`cv2.VideoCapture` errors) lack fallback options (e.g., manual login).
 - Network issues in server deployments are unhandled, requiring middleware (e.g., Flask).
- **Webcam Dependency:**
 - Cloud platforms lack webcam support, limiting deployment options.
 - A JavaScript-based webcam solution(`getUserMedia`) would require a hybrid architecture.

Technical Insights

- **Security:** Implementing JWT or OAuth for admin authentication would align with industry standards.
- **Facial Recognition:** Adding a retry mechanism or fallback authentication (e.g., OTP) could improve usability.
- **Scalability:** A microservices architecture (e.g., FastAPI for backend, Streamlit for frontend) could handle larger loads.
- **Testing:** Unit tests(`unittest` or `pytest`) for functions like `hash_password` and `cast_vote` would ensure reliability.

Significance

- Limitations highlight trade-offs for a small-scale system.
 - Addressing them would enable deployment in larger, critical elections.
 - The current design suits controlled environments (e.g., schools, organizations).
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Conclusion

The Online Voting System demonstrates the power of Python, Streamlit, SQLite, and facial recognition to create a secure, user-friendly voting platform. Its modular design, biometric authentication, and transparent result visualization make it ideal for small-scale elections. Enhanced technical details reveal opportunities for optimization, such as database performance, facial recognition accuracy, and deployment scalability. By addressing limitations like admin security and webcam dependency, the system could support broader use cases, paving the way for modern, secure e-voting solutions.